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If two fair dice are rolled, find the probability that the sum of the dice is 4, given that the sum is greater than 3.

Recall that the conditional probability of event E given event F, written $P(E|F)$, is given by the formula $P(E|F) = \frac{P(E \cap F)}{P(F)}$, where $P(F) \neq 0$. For

equally likely outcomes, $P(E|F) = \frac{n(E \cap F)}{n(F)}$.

First determine the events represented by E and F. Notice that E represents "the sum of the dice is 4."

Therefore, F represents "the sum is greater than 3."

Next, find the number of outcomes where the sum of the dice is 4 and the sum is greater than 3. This can be written as $n(E \cap F)$.

Notice that if the sum of the dice is 4, the sum is always greater than 3. Thus, find the number of events where the sum of the dice is 4, or $n(E)$.

$$n(E) = 3$$

Now find the number of events where the sum is greater than 3, or $n(F)$.

$$n(F) = 33$$

Finally, substitute the resulting values into the formula for conditional probability and simplify.

$$\begin{aligned} P(E|F) &= \frac{n(E \cap F)}{n(F)} \\ &= \frac{n(E)}{n(F)} \\ &= \frac{3}{33} \\ &= \frac{1}{11} \end{aligned}$$

Therefore, the probability that the sum of the dice is 4, given that the sum is greater than 3, is $\frac{1}{11}$.